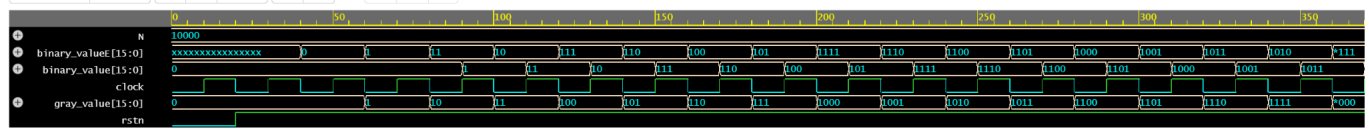
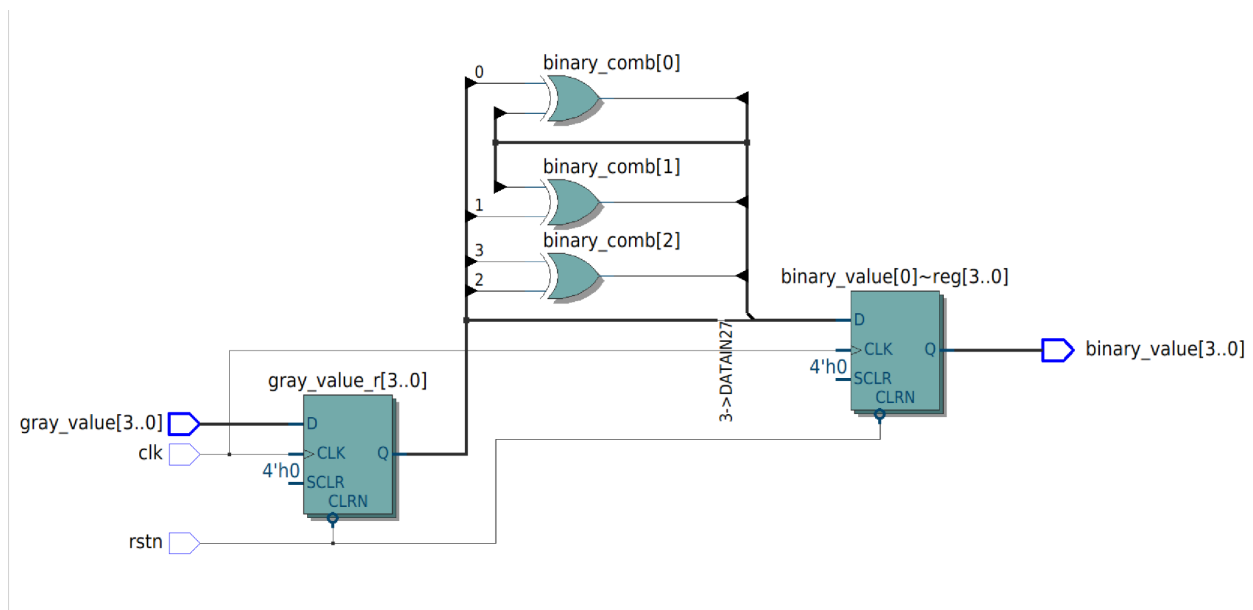


Part A



```
# gray=0000000000000000, bin_exp=0000000000000000, bin=0000000000000000 YAA!
# gray=0000000000000001, bin_exp=0000000000000001, bin=0000000000000001 YAA!
# gray=0000000000000010, bin_exp=0000000000000011, bin=0000000000000011 YAA!
# gray=0000000000000011, bin_exp=000000000000010, bin=000000000000010 YAA!
# gray=0000000000000100, bin_exp=0000000000000111, bin=0000000000000111 YAA!
# gray=0000000000000101, bin_exp=0000000000000110, bin=0000000000000110 YAA!
# gray=0000000000000110, bin_exp=0000000000000100, bin=0000000000000100 YAA!
# gray=0000000000000111, bin_exp=0000000000000101, bin=0000000000000101 YAA!
# gray=0000000000000100, bin_exp=0000000000000111, bin=0000000000000111 YAA!
# gray=0000000000000101, bin_exp=0000000000000110, bin=0000000000000110 YAA!
# gray=0000000000000110, bin_exp=0000000000000100, bin=0000000000000100 YAA!
# gray=0000000000000111, bin_exp=0000000000000101, bin=0000000000000101 YAA!
# gray=0000000000000100, bin_exp=0000000000000100, bin=0000000000000100 YAA!
# gray=0000000000000101, bin_exp=0000000000000101, bin=0000000000000101 YAA!
# gray=0000000000000110, bin_exp=0000000000000101, bin=0000000000000101 YAA!
# gray=0000000000000111, bin_exp=0000000000000101, bin=0000000000000101 YAA!
# ** Note: $stop      : testbench.sv(32)
```

YAA it works woo.

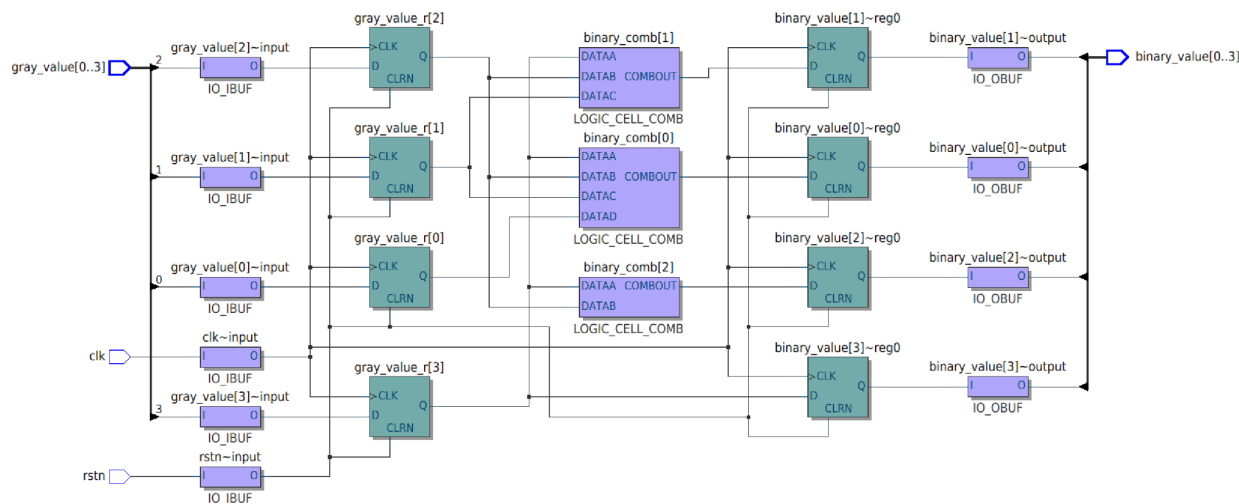


The netlist viewer clearly resembles the logic diagram given in the instructions which shows that it does model the desired behavior.

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	4
2		
3	▼ Combinational ALUT usage for logic	3
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	1
5	-- <=3 input functions	2
4		
5	Dedicated logic registers	8
6		
7	I/O pins	10
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	clk...put
12	Maximum fan-out	8
13	Total fan-out	47
14	Average fan-out	1.52

The resource usage

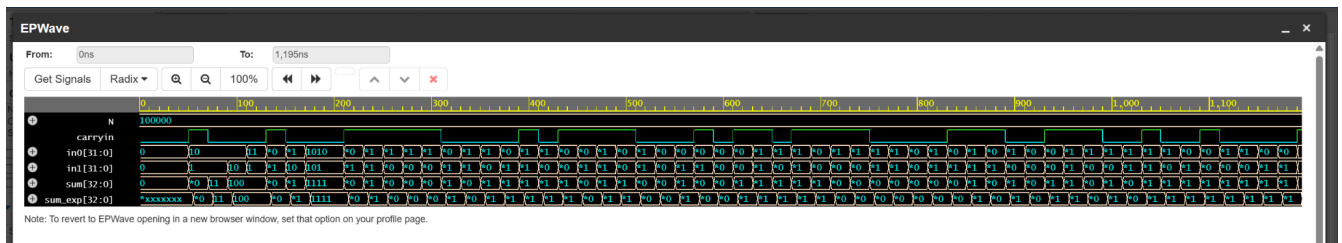
also matches what we would expect from the logic described.



	Fmax	Restricted Fmax	Clock Name	Note
1	1170.96 MHz	260.01 MHz	clk	limit due to minimum period restriction (max I/O toggle rate)

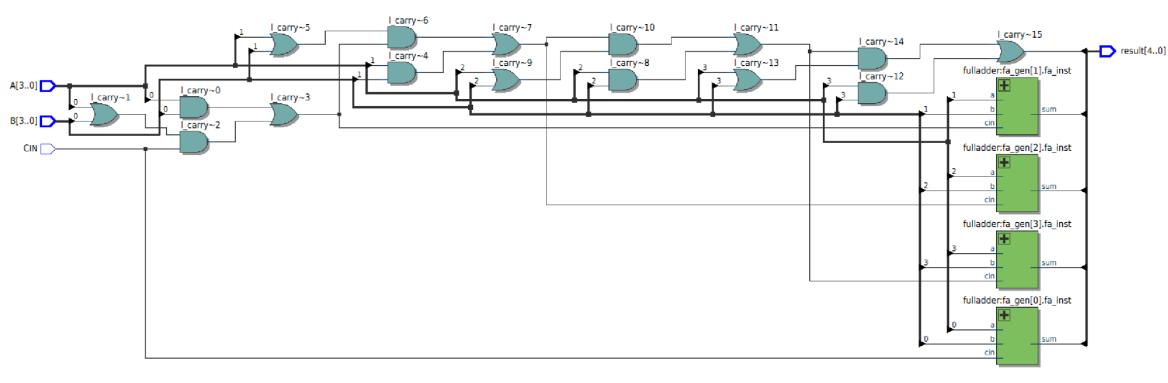
Clock rate

Part B

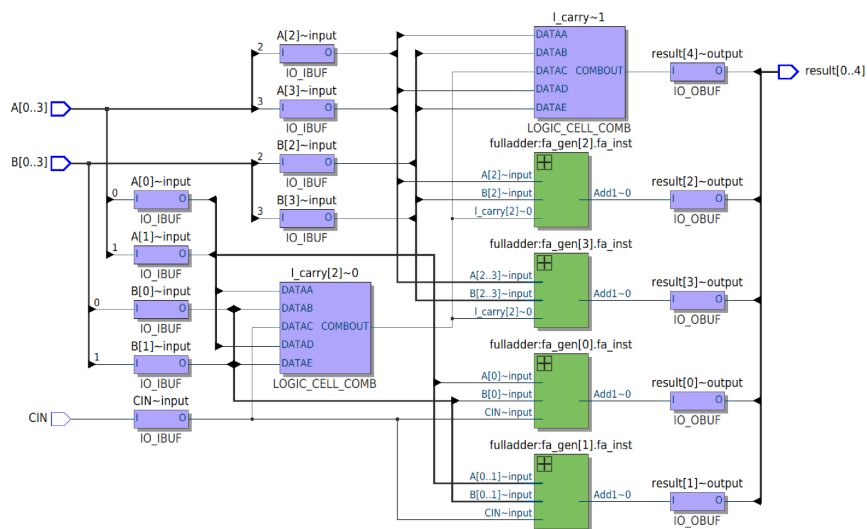


```
# time=1010 A= 349159721 B=4135772909 CIN=0 result=4484932630
#
# time=1030 A=2996811365 B=3666535093 CIN=1 result=6663346459
#
# time=1050 A=1022433657 B= 573701956 CIN=0 result=1596135613
#
# time=1070 A= 352914218 B=1442229675 CIN=0 result=1795143893
#
# time=1090 A=1851633116 B=3445537946 CIN=1 result=5297171063
#
# time=1110 A=3790668483 B= 722398550 CIN=0 result=4513067033
#
# time=1130 A=3017373287 B=2234594314 CIN=0 result=5251967601
#
# time=1150 A=2618198584 B=1020364665 CIN=0 result=3638563249
#
# time=1170 A=1249165204 B=1237736851 CIN=0 result=2486902055
#
# time=1190 A=2897726041 B=1842047451 CIN=1 result=4739773493
#
# exit
# End time: 20:30:41 on Feb 21,2026, Elapsed time: 0:00:01
# Errors: 0, Warnings: 0
```

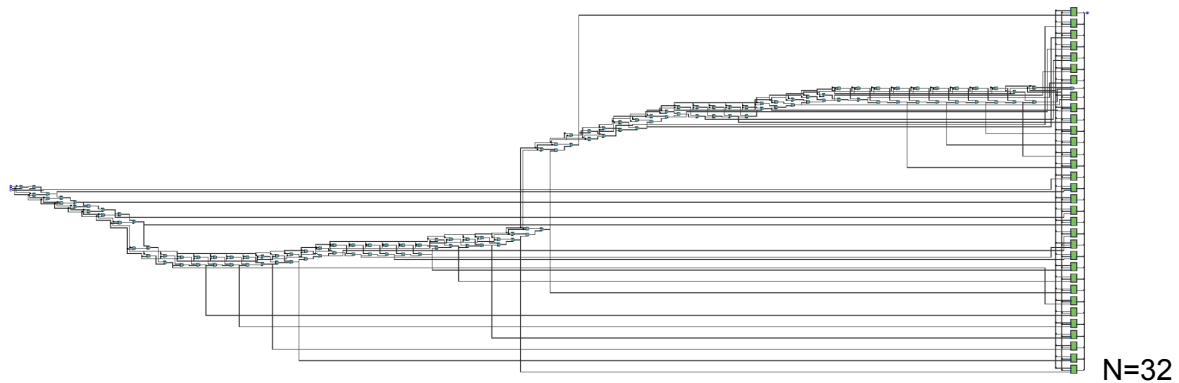
As expected from the testbench, there are no errors and the adder works as it is supposed to, with the expected value of the bits coming back correctly.



	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	3
2		
3	▼ Combinational ALUT usage for logic	6
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	4
4	-- 4 input functions	0
5	-- <=3 input functions	2
4		
5	Dedicated logic registers	0
6		
7	I/O pins	14
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	1_c...~0
12	Maximum fan-out	3
13	Total fan-out	45
14	Average fan-out	1.32

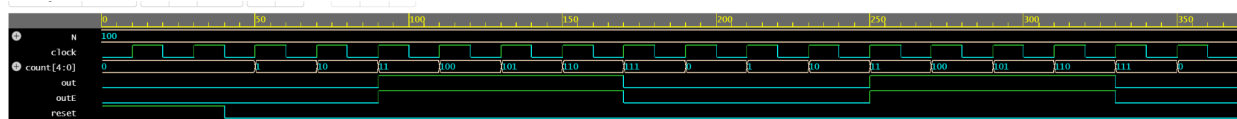


The post mapping schematic here looks pretty much exactly like the instruction hints for the carry out adder, so that's a really good sign.

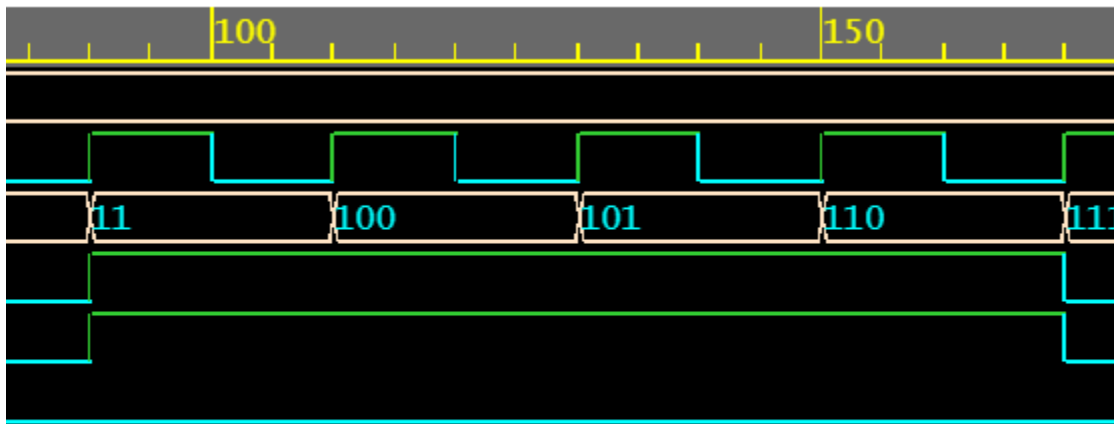


(bigger)

Part C



This one is probably best explained by the simulation here. As you can see, the N is set to 4 and the output is 4 input clock cycles:



ofile page.

Not only does this show exactly 4 clock cycles for the high signal but also that it matches exactly with the test signal... signalling how the implementation works.

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	3
2		
3	▼ Combinational ALUT usage for logic	6
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	3
4	-- 4 input functions	1
5	-- <=3 input functions	2
4		
5	Dedicated logic registers	5
6		
7	I/O pins	3
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	count[0]
12	Maximum fan-out	6
13	Total fan-out	42
14	Average fan-out	2.47

